Trustworthy AI for Cervical Cytology Screening Does Not Compose: Independent Verification of Efficiency, Domain-Shift Robustness, Uncertainty Quantification, and Explainability

Anonymous Author(s)

Submitted for double-blind review

Abstract. "Trustworthy AI" is usually evaluated as if it were one property a pipeline either has or lacks. We show instead that, for a cervical cytology screening pipeline, it decomposes into at least five independently-varying claims – whether the most parameter-efficient model is also the safest to deploy, whether any static model property predicts domain-shift robustness, whether an uncertainty-quantification method's own native confidence signal correctly flags that a model has entered a failure regime, whether combining two such signals helps, and which explainability method is most faithful – and that testing all five side by side on one fixed pipeline, rather than picking one and assuming the others follow the same pattern, is itself this paper's central contribution. First, benchmarking 12 models spanning CNN, full-size vision transformer, and efficient-vision-transformer families on SIPaKMeD cervical cytology classification, then evaluating the same models zero-shot on three independent external shift datasets (Herlev, Mendeley LBC, APCEM), we find that the two most parameter-efficient models in the registry (∼2.1M parameters each) are also the two most accurate in distribution – efficiency costs nothing here – yet the overall in-distribution winner (EfficientViT-M0) is the worst-performing model of all 12 on two of the three external datasets, dropping below the chance floor on one; no static property we measured (parameter count, FLOPs, in-distribution accuracy, architecture family) reliably predicts which model fails safely, and a CNN-family robustness pattern that survives two independent shift datasets collapses on the third. A full-registry seed-variance check (all 12 models retrained at 2 further seeds) shows this instability runs deeper than model identity: the same model can move from in-distribution winner to mid-pack shift performer under nothing more than a different random seed, so even this paper's own illustrative example should be read as one-seed evidence for a stable higher-level claim, not a reproducible model-level fact. Second, comparing four uncertainty-quantification methods' own cheapest native confidence signals on this exact backbone, method ranking itself inverts with shift severity in our original run: a five-member deep ensemble wins clearly under moderate shift, while MC Dropout's entropy is the only signal that substantially improves failure detection under severe shift – but a paired-bootstrap significance test, followed by three independent retrainings of MC Dropout under an identical recipe, shows this specific

severe-shift ranking does not reproduce (Mendeley LBC risk-coverage AUC spans 0.285–0.503 across four otherwise-identical training runs, a wider spread than separates any two *different* methods), and no method reaches a reliable abstention operating point under severe shift in either run. Third, testing explainability method choice, uncertainty- signal composition, and measured deployment cost side by side on the same pipeline: relevancy propagation wins cleanly against attention rollout and Grad-CAM on both behavioral faithfulness and real expert-annotated nucleus localization, with no condition-dependent flip, and this ranking replicates exactly on a second, independently-trained backbone; combining two uncertainty signals for selective prediction helps only under the one condition where the better individual signal was already failing, and hurts where that signal was already excellent; and resource-efficiency claims evaluated by parameter count and FLOPs alone do not reliably predict real CPU latency, while returning an explanation alongside a prediction costs roughly 3.3× a bare forward pass. Across five tested properties, three distinct resolution patterns emerge – clean and condition-independent (explainability method choice), conditional on the operating regime (uncertainty method choice, signal composition, and which specific model illustrates a given claim), and existing only once directly measured (deployment cost, architecture- family robustness patterns) – meaning a trustworthy medical AI pipeline cannot be verified with one strategy applied uniformly to every component; each property needs its own test, on its own terms, and a single seed, backbone, or shift dataset is not enough to certify any of them.

Keywords: Cervical cytology · Vision transformers · Domain shift · Model efficiency · Uncertainty quantification · Calibration · Selective prediction · Conformal prediction · Explainable AI · Trustworthy medical AI

1 Introduction

Cervical cancer remains substantially more curable when caught at a localized stage, yet screening access is documented to be uneven across rural and urban U.S. counties, and a growing literature proposes vision transformers – often marketed as "efficient" or "lightweight" variants – as classification backbones for automated cytology screening in specialist-scarce settings. A screening tool that fails silently – misclassifying confidently under conditions its training data did not cover – is more dangerous than one that fails and flags the failure, so deploying such a tool responsibly requires more than one property to hold: the backbone must be accurate *and* robust to the distribution shift it will inevitably meet across institutions, scanners, and populations; if it fails anyway, some confidence signal must correctly flag that failure; and if a clinician is to act on its output, its explanation must be faithful, not merely plausible-looking. "Trustworthy AI" is usually evaluated as if it were one property a pipeline either has or lacks. This paper's central argument is that it is not: we identify five independently-varying claims underlying a trustworthy cervical cytology screening pipeline and test

each one directly, on one fixed pipeline, rather than assuming that verifying one property licenses assuming the others.

The five claims, and how each is tested, are: (1) does a lightweight, parameter-efficient architecture trade accuracy for efficiency, tested across a 12-model registry in distribution; (2) does whichever backbone wins that in-distribution benchmark remain the safest choice under real cross-institution domain shift, tested zero-shot on three independent external datasets and, because a single seed cannot answer a question about reproducibility, re-tested across three random seeds per model; (3) does a standard uncertainty-quantification method's own native confidence signal correctly detect that a backbone has entered a shift regime where its predictions should not be trusted, tested across four methods and two shift severities, and, because a single training run cannot answer a question about a method's stability, re-tested via a significance test and repeated retraining; (4) does combining two uncertainty signals into one selective-prediction rule help, tested directly rather than assumed from either signal's individual behavior; and (5) which explainability method is most faithful, tested against both behavioral faithfulness and real expert-annotated ground truth, and re-tested on a second, independently-trained backbone. A sixth, supporting thread runs throughout: whether architectural efficiency claims and OOD-confidence-drop claims hold under direct measurement rather than assumption, which turns out to require its own verification as much as any of the five main claims.

Our contributions are organized around these five properties, plus two cross-cutting methodological points that only emerge from testing them side by side rather than in isolation. First, a 12-model efficiency/accuracy benchmark on cervical cytology classification showing that the most parameter-efficient models in the registry are also the most accurate – efficiency costs nothing here, contrary to the usual trade-off framing. Second, a systematic escalation from a 2-model to a 4-model to a full 12-model cross-dataset domain-shift evaluation across three independent external datasets, converging on: no single static model property predicts shift robustness, the in-distribution winner is the shift-robustness loser, and a robustness pattern that survives two shift datasets is not guaranteed to survive a third. Third, a full-registry seed-variance check across all 12 models at 3 random seeds each, showing that some of the paper's own illustrative single-model, single-dataset claims – including its own headline poster-child example – are seed-specific rather than general, a finding that reinforces rather than undermines the paper's central thesis by extending "no static property predicts robustness" from across models to across seeds of the same model. Fourth, a four-method uncertainty-quantification comparison on the backbone identified in contribution 2 as having a documented, severe, real distribution-shift failure, showing that method ranking itself depends on shift severity – and a significance test plus three independent retrainings showing that even the specific severe-shift ranking reported here does not reliably reproduce, a second, sharper layer of the same warning contribution 3 raises about individual models. Fifth, the first system-level test of whether combining two uncertainty signals for selective prediction helps, finding a conditional answer that extends "no single best

method" from individual methods to method combinations. Sixth, a three-way explainability comparison against both behavioral faithfulness and real expert-annotated localization ground truth, where one method wins unambiguously on every metric and replicates exactly on a second backbone – a cleaner, more stable result than any of the uncertainty-related findings, and evidence that not every property of a trustworthy pipeline is equally fragile. Seventh, a measured, not estimated, resource-efficiency and deployment-cost analysis showing that parameter count and FLOPs do not reliably predict CPU latency, and that an explanation costs $\sim$3.3$\times$ a bare forward pass. The pattern across all seven – some claims collapse under retraining, some flip with operating condition, and some hold cleanly but only once actually measured – is the paper's unifying argument: a trustworthy-AI pipeline cannot be verified with one strategy applied uniformly across its components, and each property needs its own, independent, and where possible repeated test.

2 Related Work

2.1 Efficient vision transformers

Efficiency-oriented vision transformer variants form three loose lineages relevant to the registry benchmarked here. Hybrid local–global designs (MobileViT [10], MobileViTv2 [11]) interleave convolutional stages with transformer blocks to keep early-stage compute cheap while retaining global attention at coarser resolutions. Distillation-oriented designs (DeiT [7]) instead keep a standard ViT backbone but use a distillation token and teacher network to reach competitive accuracy at far smaller pretraining-data budgets than the original ViT [5] required. Hardware-aware designs (LeViT [8], TinyViT [9]) reintroduce convolutional stages specifically to improve real-device latency, not just FLOP count, by favoring operations that map efficiently onto standard inference kernels. EfficientViT-M0 and EfficientViT-B0, the two smallest and most accurate models in this study, are frequently assumed to share a lineage because timm's tag naming groups them together – verified directly rather than assumed, they do not. EfficientViT-M0 uses cascaded group attention [13], a Microsoft-authored design that reduces redundant computation across attention heads by feeding each head a different split of the input feature map. EfficientViT-B0 belongs to an entirely unrelated MIT-authored lineage built around multi-scale linear attention for high-resolution dense prediction [12], originally designed for segmentation and detection rather than classification efficiency. That this study's two best-performing, most-efficient models come from two architecturally unrelated papers sharing only a name is itself a small caution against treating "the EfficientViT family" as one coherent design lineage in future efficiency claims. None of these efficiency- oriented design papers evaluate cross-institution domain shift as part of their own efficiency claim – efficiency and robustness are treated as separate, unrelated properties in the architecture literature this registry draws from, which is precisely the gap Sections 4.1–4.4 address.

2.2 Domain shift and model selection in medical imaging

A substantial general (not cervical-cytology-specific) medical-imaging literature documents that models selected on held-out in-distribution data can degrade sharply under real-world distribution shift across scanners, sites, and populations, motivating external-validation practices increasingly required by regulators and journals alike. However, this literature typically evaluates one or a small number of architectures rather than a systematic sweep across CNN, full-size transformer, and efficient-transformer families at matched task and data scale, which is what distinguishes this paper's 12-model, three-shift- dataset design: the question is not whether shift degrades performance (well established) but *whether any measurable static property of a model predicts which model degrades least*, tested directly rather than assumed from a single architecture's behavior.

2.3 Post-hoc calibration and its limits under shift

Temperature scaling [17] remains the standard post-hoc softmax calibration baseline: a single scalar fit on a held-out split rescales logits so that in-distribution confidence matches empirical accuracy, at essentially zero added inference cost. It is a strong in-distribution baseline (Section 4.5) but is a purely in-distribution correction by construction – it cannot introduce information the model did not already encode about out-of-distribution inputs, which is exactly why it is the weakest signal once shift severity increases (Section 4.5). This motivates comparing it against methods that inject additional signal at training or inference time rather than only recalibrating a fixed model's existing outputs.

2.4 Multi-pass and ensemble uncertainty

MC Dropout [18] approximates Bayesian inference by keeping dropout active at test time and treating the spread across repeated stochastic forward passes as an uncertainty signal, at a cost proportional to the number of passes. Deep ensembles [19] instead train several independently- initialized models and use their disagreement, at a cost proportional to ensemble size; empirically, ensemble diversity has been argued in the broader literature to capture a different, complementary source of uncertainty (functional diversity across independently-trained models) than a single model's own stochastic-forward-pass variance, which is consistent with this paper's finding (Section 4.5) that the two methods are not redundant with each other – they win under different shift severities rather than one strictly dominating the other.

2.5 Single-pass uncertainty and evidential learning

Evidential deep learning [20] reframes classification as Dirichlet evidence accumulation, producing a single-pass uncertainty signal (vacuity) without the multi-pass or multi-model inference cost MC Dropout and deep ensembles incur – an attractive property for latency-constrained deployment, and the reason it is

included here despite calibrating markedly worse than the other three methods in distribution (Section 4.5). Whether that single-pass efficiency advantage is worth its calibration cost is precisely an empirical, task-dependent question this comparison is designed to answer rather than assume from the method's design motivation alone.

2.6 Uncertainty quantification in cervical cancer screening

Adjacent work applies conformal prediction and Monte Carlo inference specifically to cervical cancer screening uncertainty, typically proposing or evaluating one uncertainty method in isolation against a single baseline. This paper differs by comparing four methods head-to-head on a documented real (not assumed or synthetically induced) shift-failure case, by evaluating each method across two shift severities rather than one – which is precisely what reveals the rank inversion in Section 4.5 a single-severity evaluation could not have surfaced – and, further, by testing whether that inversion itself survives retraining (Section 4.6), a question adjacent single-run comparisons in this literature do not typically ask of themselves.

2.7 Transformer explainability methods

Attention rollout [22] treats a transformer's own attention weights, chained across layers via a fixed recurrence, as a relevance signal, on the premise that attention indicates where a model "looks." This premise has itself been contested in the broader literature examining whether attention weights are faithful explanations of a model's actual decision process rather than merely correlated with it – a concern this paper's own faithfulness metrics (Section 4.7) are designed to test empirically rather than assume either way. Relevancy propagation [21] extends attention rollout by weighting each transformer block's attention matrix with its gradient with respect to the target class before the same rollout aggregation, making the resulting relevance map class-discriminative rather than a fixed structural property of attention, and is the primary method evaluated here specifically because it addresses the class-agnosticity critique of raw attention directly. Grad-CAM [23] was originally designed for CNN feature maps, where a single late convolutional layer naturally encodes class-relevant spatial structure; its adaptation to transformers by targeting a block's token-wise activations instead of a convolutional feature map is common practice in the literature, but the specific failure mode we identify and document (Section 3.6) for CLS-token-pooled ViTs – an all-zero heatmap if the wrong block is targeted – does not appear to be widely flagged in prior adaptations, despite being silent rather than an error.

2.8 Selective prediction and conformal calibration

Split conformal prediction provides a distribution-free, finite-sample coverage guarantee from a single calibration split, at the cost of producing a set of plausible labels rather than one score; its prediction set size is a natural, model-agnostic

uncertainty proxy, but one that is coarse (a small integer count) compared to a continuous confidence score. This paper's system-integration question – whether combining temperature-scaled confidence with conformal set size outperforms either alone (Section 4.9) – is, to our knowledge, not directly addressed by prior work proposing either signal independently, since each is typically evaluated as a self-contained method rather than as one input to a combined abstention rule.

2.9 Faithfulness evaluation

Deletion/insertion faithfulness [24] evaluates an explanation by its behavioral effect on the model itself (does removing or revealing the patches an explanation calls important actually change the model's prediction), rather than by agreement with a human-annotated region, and is the behavioral-faithfulness metric used throughout Section 4.7. Evaluating localization against SIPaKMeD's own expert-marked nucleus polygons, rather than a segmentation model's derived output, avoids compounding this paper's explainability evaluation with a second model's own errors.

2.10 Closely related prior work

Directly adjacent to this paper's own scope: a stacking-ensemble approach combined with explainable AI for cervical cancer diagnosis [25], which this paper differs from by evaluating explainability methods against a real expert-annotated localization ground truth and behavioral faithfulness metrics, and by evaluating efficiency–robustness trade-offs across a full 12-model architecture- family registry, rather than an ensemble-accuracy framing over a single model; and a lightweight-ViT-plus-explainability design for plant disease diagnosis, "MaizeFormerX" [26], which motivates this paper's own choice to hold efficiency and explainability as joint, not competing, design goals. Two further papers sit in adjacent clinical-AI territory: a semi-supervised retinal disease recognition system [27] and an explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support [28], both part of a broader body of work in trustworthy, explainable clinical AI of which this paper's cervical cytology pipeline is one instance.

3 Methods

3.1 Overview: which backbone answers which question

Three of this paper's five properties (efficiency, shift robustness, and its seed-variance check) are tested across the full 12-model registry (Section 3.3), since the question there is precisely whether backbone identity predicts an outcome, which requires varying backbone identity. The uncertainty-quantification comparison (Section 3.5) instead holds one backbone fixed – EfficientViT-M0, the registry's in-distribution winner and, per Section 4.2, its worst-performing model under

two of three external shifts – to isolate the uncertainty-quantification method rather than architecture, and because this is the exact backbone for which the question "can a confidence signal flag this specific failure" is most practically urgent. The explainability comparison (Section 3.6) uses a third backbone, DeiT-Tiny, instead: relevancy propagation and attention rollout are both defined only for standard multi-head self-attention transformer blocks, and EfficientViT-M0's hybrid convolutional + cascaded-group-attention design has no established propagation rule for either method, so forcing an ad hoc extension onto an architecture these methods were not designed for was avoided in favor of a standard `timm VisionTransformer` (DeiT-Tiny, 0.9606 balanced accuracy on the same in-distribution task, vs. EfficientViT-M0's 0.9654) retrained fresh for this purpose. The explainability ranking is re-tested on a second, independently-trained backbone of the same family (ViT-Small, Section 4.8) to confirm it is not one lucky model's property; the system-integration and resource-efficiency results (Sections 4.9–4.10) reuse DeiT-Tiny and the Herlev/Mendeley LBC shift conditions established in Section 3.4.

3.2 Datasets

SIPaKMeD [1] (4,049 native-resolution single-cell crops derived directly from the dataset's own cytoplasm-polygon boundary annotations, not a pre-resized third-party mirror) is the in-distribution training and test dataset throughout, with 5 Bethesda-relevant cytomorphology classes. Three independent external datasets provide zero-shot cross-institution shift evaluation, harmonized onto one shared binary normal/abnormal label: **Herlev** [14] (moderate shift), **Mendeley LBC** [15] (severe shift), and the **APCEM-Enhanced Cervical Cytology Pap Smear Image Dataset** [16] (4,253 images across the 6-class Bethesda taxonomy – NILM, ASC-US, ASC-H, LSIL, HSIL, SSC – harmonized identically: NILM normal, all five atypical/dysplastic classes abnormal). APCEM differs from Herlev and Mendeley LBC in acquisition pipeline, institution, and label granularity, so it functions as a genuinely independent replication of the domain-shift evaluation rather than a second look at the same shift.

3.3 Model registry

Twelve models spanning three families: CNN baselines (ResNet-50 [2], MobileNetV2 [3], EfficientNet-Lite0 [4]), full-size vision transformers (ViT-Small [5], Swin-Tiny [6]), and efficient/mobile-oriented vision transformers (DeiT-Tiny [7], LeViT-128 [8], TinyViT-5M [9], MobileViT-S [10], MobileViTv2 [11], EfficientViT-B0 [12], EfficientViT-M0 [13]). All timm tags verified against the pinned timm version rather than assumed from memory; the two EfficientViT variants' source papers were confirmed independently against timm's own model cards rather than assumed from the shared tag prefix (Section 2.1).

3.4 Training and in-distribution evaluation

All 12 models trained on SIPaKMeD, 60/15/10/15 train/val/calibration/test split, seed 13 as the headline seed (extended to seeds 17 and 29 for the seed-variance check, Section 4.4), full 30-epoch budget with early stopping (patience 7), on a single GPU. Reported: balanced accuracy, macro F1, AUROC, parameter count, and measured (not estimated) inference latency.

3.5 Cross-dataset domain-shift evaluation

Each model evaluated zero-shot – no retraining, no recalibration – on Herlev, Mendeley LBC, and APCEM after harmonizing all datasets' differing label taxonomies onto one shared binary normal/abnormal label. This tests generalization under the conditions this application will actually face: different institutions, scanners, and staining protocols, not a synthetic corruption of the same source images. The seed-variance check (Section 4.4) repeats this protocol for all 12 models at 2 further seeds, training once per model per seed and evaluating the same checkpoint on all three external datasets, rather than retraining separately per dataset.

3.6 Uncertainty-quantification methods

Four methods, each using its own cheapest native confidence signal (a single native signal, not a method-specific post-hoc correction layered on top, so the comparison reflects what a deployed system would actually have access to without extra engineering), all sharing one manifest, one calibration split, and one backbone initialization recipe (EfficientViT-M0): (1) **Baseline**: cross-entropy training, post-hoc temperature scaling fit on a held-out calibration split; confidence = temperature- scaled max-softmax probability. (2) **MC Dropout**: dropout rate 0.3, 20 stochastic forward passes at inference; confidence = negative predictive entropy of the mean prediction across passes. (3) **Evidential deep learning**: Dirichlet evidence output via a softplus activation and the standard evidential loss (Type II MLE plus an annealed KL-divergence-to-uniform penalty); confidence = 1 − vacuity. (4) **Deep ensemble**: 5 independently-seeded members, each trained with the same recipe as the baseline; confidence = negative variance of the ensemble's mean predicted probability across members.

A methods-level fix was required before any GPU time was spent on this comparison: the backbone's only `nn.Dropout` module ships at $p = 0$ by default in its pretrained configuration – verified directly via model inspection, not assumed – which would make MC Dropout completely degenerate (zero stochasticity across passes). A dropout rate is forced to a real, non-zero value before training begins, not just patched in at inference time, since MC Dropout's Bayesian-approximation justification specifically requires dropout to be active during training as well as at test time. We flag this because the failure mode is silent (MC Dropout would simply produce zero-variance, falsely confident "uncertainty" estimates) rather than an error.

All four methods evaluated identically on three splits: the SIPaKMeD in-distribution test set, Herlev (moderate shift), and Mendeley LBC (severe shift). Metrics: expected calibration error (ECE), full-coverage accuracy, risk-coverage AUC (integral of error rate over the coverage curve – lower is better, since a well-behaved confidence signal keeps risk near zero until coverage is forced high), and selective accuracy at fixed coverage levels.

3.7 Explainability methods

All three methods produce a relevance map over the backbone's 14×14 patch grid (DeiT-Tiny, 224px input, 16px patches), upsampled to image resolution. **Attention rollout**: raw softmax attention, averaged over heads, chained across all 12 blocks via the standard rollout recurrence. Class-agnostic. **Relevancy propagation**: identical rollout recurrence, but each block's contribution is $\mathrm{ReLU}(\nabla_{\mathrm{attn}} \odot \mathrm{attn})$, averaged over heads, where the gradient is with respect to the model's own predicted class's logit. Class-discriminative. **Grad-CAM (ViT-adapted)**: gradient-weighted channel importance applied to a transformer block's token activations rather than a CNN's convolutional feature map. *Implementation note*: the target layer must be the *second-to-last* transformer block, not the last. DeiT-Tiny pools only the CLS token for classification, so no layer downstream of the final block mixes information back into patch-token positions – verified directly by inspecting gradients: every patch token's gradient at the last block's output was exactly zero, while the second-to-last block's patch-token gradients vary normally.

Deletion/insertion AUC: patches are ranked by a method's own relevance score and progressively replaced with (deletion) or revealed from (insertion) a heavily Gaussian-blurred version of the image, tracking the model's predicted-class probability at each step; the area under the resulting probability-vs-fraction curve is reported (lower is better for deletion, higher is better for insertion). **Nucleus concentration ratio**: SIPaKMeD ships real, expert-marked nucleus boundary polygons per cell; we persist these in crop-local pixel coordinates and compute the fraction of a heatmap's total relevance mass falling inside the nucleus region, divided by the nucleus's share of image area. A ratio of 1.0 is exactly chance level; higher values mean a heatmap concentrates relevance in the nucleus more than chance would predict – clinically the correct place to concentrate.

3.8 Calibration, conformal prediction, and the combined selective-prediction signal

Temperature scaling is fit on a held-out calibration split disjoint from both training and test data. Conformal prediction sets are built via a resampling-averaged split-conformal threshold. The combined selective-prediction signal rank-normalizes temperature-scaled max-softmax confidence and (inverted) conformal prediction-set size independently, then averages their ranks. Two safe-composition subclaims hold by construction given this project's post-hoc-attachment architecture, and were verified rather than assumed: temperature scaling changed **zero**

argmax predictions on the calibration split, and explainability computation is a read-only analysis that cannot touch model weights.

3.9 Efficiency and latency measurement

Parameters and FLOPs (via `torch.utils.flop_counter.FlopCounterMode`) are architecture properties independent of trained weight values, so measured without any training. CPU latency and memory (isolated per model in its own subprocess) are measured with untrained weights for the same reason. The "cost of an explanation" is measured as a full forward+backward+rollout pass for relevancy propagation's own predicted-class explanation, against a bare forward pass, both on CPU.

4 Results

4.1 Efficiency/accuracy benchmark

Table 1. Efficiency/accuracy benchmark, all 12 models, sorted by balanced accuracy.

Model	Bal. acc.	Macro F1	AUROC	Params	Latency (ms)
EfficientViT-M0	**0.9654**	**0.9656**	0.9985	**2,157,401**	17.56
EfficientViT-B0	0.9623	0.9624	0.9983	2,135,413	6.97
DeiT-Tiny	0.9606	0.9607	0.9974	5,525,381	5.17
Swin-Tiny	0.9605	0.9606	0.9965	27,523,199	11.36
TinyViT-5M	0.9587	0.9589	0.9979	5,073,369	9.57
MobileViT-S	0.9570	0.9575	0.9979	4,940,837	9.32
ResNet-50	0.9542	0.9540	0.9966	23,518,277	5.95
LeViT-128	0.9458	0.9456	0.9946	8,447,786	12.81
MobileViTv2	0.9387	0.9401	0.9966	4,391,406	9.90
EfficientNet-Lite0	0.9375	0.9376	0.9944	3,377,413	5.51
ViT-Small	0.9374	0.9380	0.9960	21,667,589	5.39
MobileNetV2	0.9357	0.9365	0.9952	2,230,277	7.01

The two smallest models in the registry are also the two best-performing, beating every full-size baseline (ResNet-50, 23.5M params; Swin-Tiny, 27.5M params) by 1–3 points of balanced accuracy. On this task, at this data scale, the efficiency-accuracy trade-off does not appear at all in-distribution – whether this generalizes is exactly what Section 4.2 tests, and the answer is not reassuring.

4.2 Cross-dataset domain shift, escalating from 2 to 12 models

Stage 1 (DeiT-Tiny alone): in distribution 0.9504 balanced accuracy, Herlev (zero-shot) 0.7151, Mendeley LBC (zero-shot) 0.5230 – a model that looks excellent in distribution degrades to near-chance on the more severe external dataset, with nothing in a standard pipeline able to flag this.

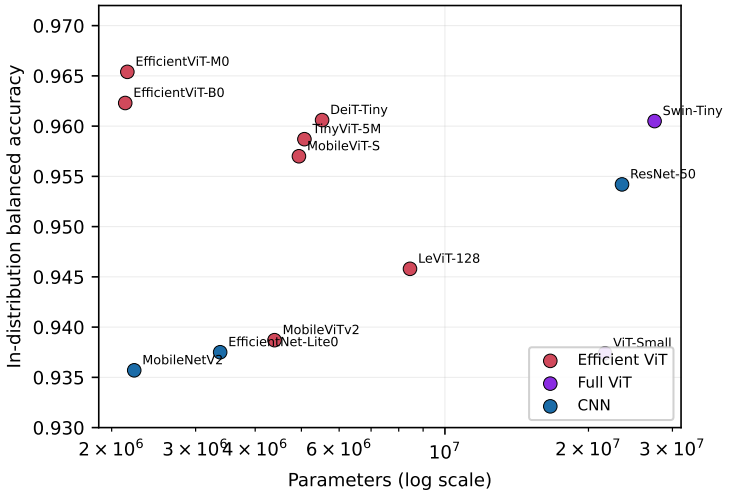


Fig. 1. Parameters (log scale) vs. in-distribution balanced accuracy, colored by architecture family. The two efficient-ViT models with the fewest parameters (EfficientViT-M0/B0) are also the two most accurate.

Stage 2 (adding EfficientViT-M0): the *better* in-distribution model (0.9737 vs. DeiT-Tiny's 0.9504) is the *worse* model under both shifts (Herlev 0.6716 vs. 0.7151; Mendeley LBC 0.4731 vs. 0.5230 – the first model in this study to drop below the 0.50 chance floor). Two models this close in architecture family and parameter count producing this reversal is a direct caution against using in-distribution accuracy as a proxy for deployment robustness.

Stage 3 (adding ResNet-50 and Swin-Tiny):

Table 2. One model per architecture family under cross-dataset shift.

Model (family)	SIPaKMeD	Herlev	Mendeley LBC
EfficientViT-M0 (efficient ViT)	**0.9737**	0.6716	**0.4731** (below chance)
Swin-Tiny (full ViT)	0.9590	**0.7282**	0.5192
DeiT-Tiny (efficient ViT)	0.9504	0.7151	0.5230
ResNet-50 (CNN)	0.9556	0.6838	**0.6082**

Not a clean CNN-vs-transformer split: on Herlev, both transformer models beat ResNet-50. What holds across both shift datasets is narrower: the single most parameter-efficient, best-in-distribution model (EfficientViT-M0) is the worst performer under shift on both external datasets, and the only one below chance on either.

Stage 4 (all 12 models):

Table 3. Full 12-model cross-dataset shift evaluation, sorted by Mendeley LBC score.

Model (family, params)	SIPaKMeD	Herlev	Mendeley LBC
EfficientNet-Lite0 (CNN, 3.38M)	0.9455	0.5970	**0.6674**
ResNet-50 (CNN, 23.5M)	0.9556	0.6838	0.6082
MobileViTv2 (eff. ViT, 4.39M)	0.9638	**0.7501**	0.5418
MobileNetV2 (CNN, 2.23M)	0.9505	0.6919	0.5391
DeiT-Tiny (eff. ViT, 5.5M)	0.9504	0.7151	0.5230
EfficientViT-B0 (eff. ViT, 2.14M)	0.9738	0.6818	0.5217
Swin-Tiny (full ViT, 27.5M)	0.9590	0.7282	0.5192
ViT-Small (full ViT, 21.7M)	0.9573	0.7049	0.5084
LeViT-128 (eff. ViT, 8.4M)	0.9526	0.5777	0.4764
EfficientViT-M0 (eff. ViT, 2.16M)	0.9737	0.6716	0.4731
TinyViT-5M (eff. ViT, 5.1M)	0.9685	**0.7327**	0.4543
MobileViT-S (eff. ViT, 4.94M)	0.9520	0.7104	**0.2927** (AUROC 0.215)

A near-clean CNN-vs-transformer split emerges on Mendeley LBC, with one exception: all three CNNs land at 0.539–0.667; eight of nine transformer-family models land at 0.293–0.523; the ninth (MobileViTv2) scores 0.5418, above the lowest CNN (0.5391) by 0.003 – a gap seed-variance testing (Section 4.4) shows is noise, not a real difference. This split does not hold on Herlev, where transformer models occupy both the best (MobileViTv2, TinyViT-5M) and weaker (LeViT-128) positions.

MobileViT-S is a severe, singular outlier: its Mendeley LBC AUROC (0.215) is not merely below chance but actively anti-correlated with ground truth. Its direct successor, MobileViTv2, shows no such problem and is the single strongest model in the entire registry on Herlev (0.7501) – the same architecture-family name spanning a catastrophic failure and a top performer, ruling out architecture lineage as a predictor of shift robustness as thoroughly as parameter count and in-distribution accuracy already have been.

4.3 A third, independent shift dataset (APCEM)

APCEM is a genuinely independent replication – different acquisition pipeline, institution, and label granularity from either Herlev or Mendeley LBC – and it does not merely confirm the earlier finding that in-distribution accuracy fails to predict shift robustness (11 of 12 models land at or below the 0.50 chance floor for balanced accuracy on APCEM, several with AUROC well below 0.35), it overturns the one pattern that had looked stable across Stages 3–4: that CNN-family models cluster above chance while transformer-family models cluster at or below it. Here the CNN family alone spans the single best-performing model in the entire registry (MobileNetV2, 0.6158 balanced accuracy, the only model of the 12 to clear chance meaningfully) and one of the single worst (ResNet-50, AUROC 0.2929, second-lowest of all 12) – the same family that was the more robust one, on average, on Mendeley LBC. DeiT-Tiny and EfficientViT-M0, the

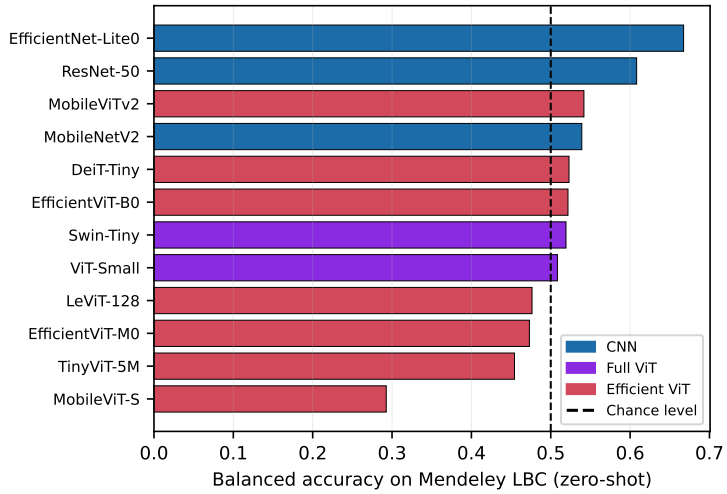


Fig. 2. Balanced accuracy on Mendeley LBC (zero-shot), sorted, colored by architecture family. All three CNNs cluster above chance; eight of nine transformer-family models cluster at or below chance, with MobileViT-S the sole catastrophic outlier.

Table 4. Full 12-model evaluation on the third external shift dataset (APCEM), sorted by AUROC.

Model (family, params)	SIPaKMeD	APCEM bal. acc.	APCEM AUROC
MobileNetV2 (CNN, 2.23M)	0.9623	**0.6158**	**0.6528**
EfficientNet-Lite0 (CNN, 3.38M)	0.9309	0.5393	0.5701
LeViT-128 (eff. ViT, 8.4M)	0.9001	0.5263	0.5376
Swin-Tiny (full ViT, 27.5M)	0.9330	0.4724	0.3082
EfficientViT-B0 (eff. ViT, 2.14M)	0.9638	0.4481	0.3206
DeiT-Tiny (eff. ViT, 5.5M)	0.9589	0.4469	0.3799
MobileViTv2 (eff. ViT, 4.39M)	0.9570	0.4290	0.4120
TinyViT-5M (eff. ViT, 5.1M)	0.9670	0.4100	0.3681
EfficientViT-M0 (eff. ViT, 2.16M)	0.9621	0.3730	0.3407
MobileViT-S (eff. ViT, 4.94M)	0.9687	0.3621	0.3161
ViT-Small (full ViT, 21.7M)	0.9538	0.3444	0.2555
ResNet-50 (CNN, 23.5M)	0.9639	**0.3339**	**0.2929**

two models directly compared since Stage 2, again replicate their earlier ranking (DeiT-Tiny 0.4469 vs. EfficientViT-M0 0.3730), with EfficientViT-M0 again the weaker model under shift despite again being the stronger in-distribution model on this run (0.9621 vs. 0.9589) – the same reversal, now observed on a third, independently sourced dataset.

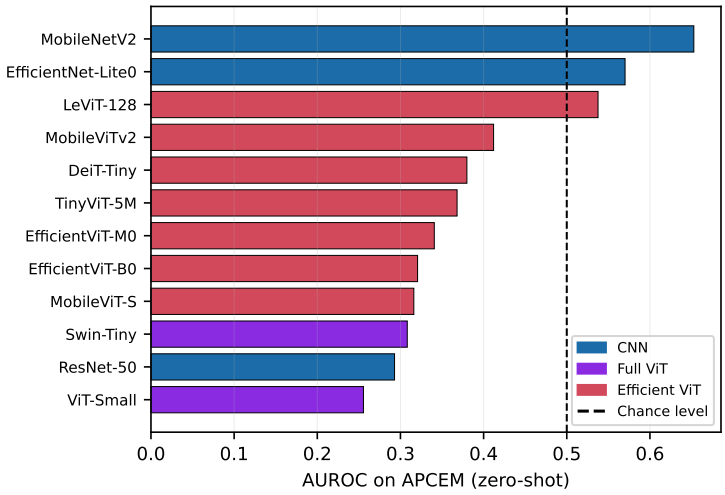


Fig. 3. AUROC on APCEM (zero-shot), sorted, colored by architecture family. Unlike Mendeley LBC, the CNN family spans both the single best (MobileNetV2) and one of the single worst (ResNet-50) models in the registry – no family-level pattern survives this third dataset.

4.4 Full-registry seed-variance check: how much of the above is one seed's story?

Every result above is a single train/val/calibration/test split (seed 13). To test whether the paper's own headline patterns are properties of the models and datasets, or artifacts of that one seed, we retrained all 12 models at 2 further seeds (17, 29), one training run per model per seed evaluated zero-shot on all three external datasets from the same checkpoint, and combined this with the original seed-13 numbers above.

The Mendeley LBC "near-clean CNN-vs-transformer split" is itself seed-fragile. At seed 13, only 1 of 9 transformer-family models (MobileViTv2, by 0.003) exceeds the weakest CNN. At seed 17, 3 of 9 do, by margins up to 0.098 (EfficientViT-B0). At seed 29, 4 of 9 do, by margins up to 0.113. The margins at seeds 17 and 29 are an order of magnitude past the single close call already flagged as noise at seed 13 – the "near-clean split" description of Mendeley LBC

(Section 4.2, Stage 4) is a property of that specific seed, not a reproducible property of the dataset or the model families.

Herlev's inverse pattern (transformers beating CNNs) is, by contrast, seed-stable: 8, 7, and 8 of 9 transformer models exceed the weakest CNN at seeds 13, 17, and 29 respectively. Not every seed-checked pattern is fragile.

MobileNetV2's APCEM win is seed-stable: it is the best-performing model on APCEM at every seed tested (rank 1, 2, and 1 of 12 at seeds 13, 17, 29) – a genuinely reproducible result, not a single-seed artifact.

EfficientViT-M0's specific poster-child reversal is not. This paper's running illustration of "the in-distribution winner is the shift-robustness loser" (Sections 1, 4.2) does not reliably occupy either role across seeds:

Table 5. EfficientViT-M0's in-distribution and shift ranks (of 12) across 3 seeds.

Seed	In-distribution rank	Herlev shift rank	Mendeley LBC shift rank
13	2	10	10
17	10	7	7
29	1	5	4

At seed 13, EfficientViT-M0 is (virtually tied for) the in-distribution winner and one of the worst shift performers on both external datasets – exactly the reversal illustrated in Section 1. At seed 29, it is the outright in-distribution winner *and* mid-pack under shift on both datasets (rank 4–5 of 12) – no reversal at all. At seed 17 it is not even the in-distribution winner (rank 10 of 12). This does not weaken the paper's central claim; if anything it reinforces it. The general claim this example illustrates – no static property predicts shift robustness – is a further instance of exactly that unreliability, now shown across seeds of the same model rather than only across different models. What changes is that this specific model should be read as a seed-13-specific illustration of the general point, not a reproducible property of EfficientViT-M0 itself.

4.5 Uncertainty-quantification comparison

Table 6. Four uncertainty methods, in distribution and under two external shifts.

Method	In-dist.		Herlev		Mendeley LBC		
	ECE	Acc.	RC-AUC	Acc.	RC-AUC	Acc.	Sel. acc.@90%
Baseline (temp.)	0.012	0.967	0.193	0.694	0.515	0.530	0.522
MC Dropout	0.013	0.972	0.203	0.658	**0.285**	0.579	**0.583**
Evidential	0.159	0.967	0.167	0.677	0.476	0.543	0.539
Deep ensemble	0.012	0.974	**0.147**	**0.771**	0.473	0.537	0.544

In distribution, three methods are close; evidential is the outlier. Baseline, MC Dropout, and the deep ensemble all calibrate well (ECE 0.012–0.013) with full-coverage accuracy in the 0.967–0.974 range. Evidential deep learning's raw belief masses are markedly less calibrated (ECE 0.159) at matched point accuracy (0.967) – its probabilities would need their own post-hoc correction, deliberately not applied here, to make this comparison's calibration numbers fair to it.

Under moderate shift (Herlev), the deep ensemble wins clearly. Its risk-coverage AUC (0.147) and full-coverage accuracy (0.771) both beat the other three methods by a real margin – member disagreement is doing genuine, useful work at this shift severity.

Under severe shift (Mendeley LBC) – the condition that matters most, since it is where this exact backbone is known to fail – the ranking inverts. MC Dropout's entropy is the only signal showing a substantial, unambiguous improvement over the uncalibrated baseline (risk-coverage AUC 0.515 to 0.285; full-coverage accuracy 0.530 to 0.579). Evidential (0.476) and the deep ensemble (0.473) are both only marginally better than the plain baseline – essentially within noise of each other – despite the ensemble's clear win on the milder shift one row above. Member disagreement, which correctly flags moderate shift, does not reliably track the specific failure mode behind the severe collapse.

4.6 A significance test, and an unexpected non-replication

To test whether MC Dropout's apparent Mendeley LBC advantage over the deep ensemble (0.285 vs. 0.473) is distinguishable from bootstrap sampling noise, we ran a paired bootstrap test (2000 resamples) on per-sample risk-coverage curves. Because only Section 4.5's aggregate metrics, not its per-sample predictions, had been retained, this required retraining MC Dropout from scratch – with the identical seed (13), dropout rate, epoch budget, and manifest used above – while the deep ensemble was reloaded from its five original saved checkpoints. The retrained MC Dropout model did not reproduce the earlier result: its Mendeley LBC risk-coverage AUC came out at 0.503, not 0.285, despite every controllable factor being held fixed. Against this retrained run, the paired bootstrap test found no statistically significant difference from the deep ensemble (0.503 vs. 0.474; 95% CI for the difference $[-0.066, 0.010]$, one-sided $p = 0.929$). This is a stronger and more concerning result than a mere failure to reach significance: a fixed random seed did not reproduce a result on this specific severe-shift metric, and the gap between the two training runs of the identical MC Dropout recipe (0.285 vs. 0.503) is larger than the sampling uncertainty the bootstrap places around either individual run. To see whether these two runs were themselves a coincidence, we retrained MC Dropout twice more at genuinely different seeds (17, 29, otherwise identical recipe): Mendeley LBC risk-coverage AUC came out at 0.358 and 0.359 – both landing squarely inside the 0.285–0.503 range the first two runs staked out, rather than clustering near either endpoint. Four independent training runs of one fixed method on one fixed backbone now span

0.285 to 0.503, with a spread (0.218) far larger than the 0.147–0.203 gap that separated any two *different* methods in Section 4.5's Herlev column. Read together, the four runs mean the reported severe-shift rank inversion is a property of one specific trained model, not a property of "MC Dropout" as a method on this backbone – retraining alone, with no change to data, architecture, or hyperparameters, was sufficient to erase the reported advantage entirely, and the method's severe-shift behavior is not concentrated near any single value at all.

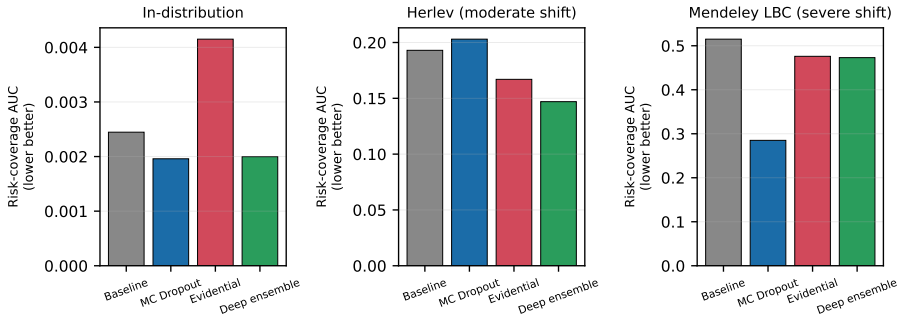


Fig. 4. Risk-coverage AUC (lower is better) per method, at three conditions with independent y-axes (note the ∼100× scale difference between in-distribution and Mendeley LBC). The best method in this run changes from deep ensemble (Herlev) to MC Dropout (Mendeley LBC), though a retraining check found the Mendeley LBC advantage does not reproduce (see text).

No method reaches a reliable abstention operating point under severe shift. Even MC Dropout's best result – 0.583 selective accuracy at 90% coverage on a binary task – is barely better than chance after abstaining on the least-confident decile. We report this plainly rather than treating "best of four" as equivalent to "good enough": for deployment purposes, none of these four methods alone would currently justify an "abstain and refer" policy on a shift this severe, in either run.

4.7 Explainability comparison

Relevancy propagation wins every comparison, and by a wide margin on nucleus concentration specifically (2.71× chance vs. attention rollout's 1.18× – barely above chance). We attribute this to a mechanistic difference rather than measurement noise: attention rollout is class-agnostic, an inherent disadvantage on an inherently class-discriminative faithfulness test, while relevancy propagation is both class-discriminative *and* integrates the gradient-weighted signal across all twelve transformer blocks.

Table 7. Explainability method comparison, 50 SIPaKMeD test images, all matched to a real nucleus polygon.

Method	Deletion AUC↓	Insertion AUC↑	Nucleus conc.↑ (1.0 = chance)
Relevancy propagation	**0.734**	**0.970**	**2.714**
Grad-CAM (ViT-adapted)	0.818	0.877	1.472
Attention rollout	0.825	0.936	1.177

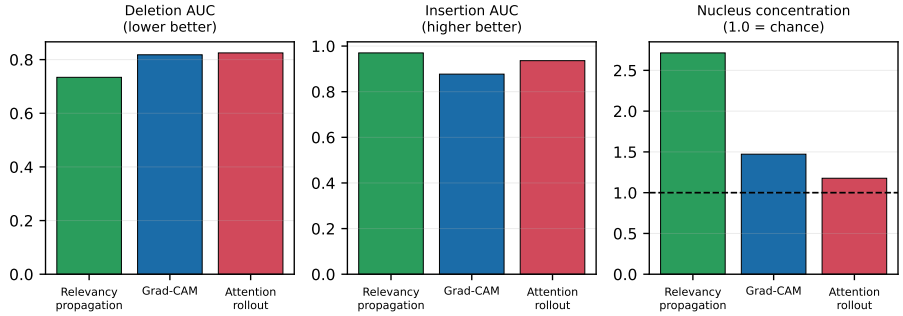


Fig. 5. Deletion AUC, insertion AUC, and nucleus concentration ratio for the three explainability methods. Relevancy propagation wins on all three, with the widest margin on nucleus concentration.

4.8 A second backbone: does the explainability ranking generalize?

Relevancy propagation's clean, unconditional win is the one result in this paper that looked, on its face, like it might not need condition-by-condition re-testing – but that claim rests on a single backbone (DeiT-Tiny), so we re-ran the identical comparison on ViT-Small, a structurally identical `VisionTransformer` (same `global_pool="token"`, `num_prefix_tokens=1`, 12 blocks – verified directly, not assumed) trained fresh with the same recipe and seed.

Table 8. Explainability ranking on a second backbone (ViT-Small) vs. the original (DeiT-Tiny).

	DeiT-Tiny (original)			ViT-Small (2nd backbone)		
Method	Del. AUC↓	Ins. AUC↑	Conc.↑	Del. AUC↓	Ins. AUC↑	Conc.↑
Relevancy prop.	**0.734**	**0.970**	**2.71**	**0.816**	**0.965**	**4.20**
Grad-CAM	0.818	0.877	1.47	0.888	0.895	1.35
Attn. rollout	0.825	0.936	1.18	0.888	0.927	0.97

The ranking replicates exactly: relevancy propagation wins all three metrics on ViT-Small too, with an even wider nucleus-concentration margin (4.20×

chance vs. its closest competitor's 1.35×) than on DeiT-Tiny. This is the one property in this paper's tested set that generalizes cleanly across a backbone change without qualification – in contrast to Section 4.9's uncertainty-composition result, which flips direction depending on shift severity, and Section 4.10's efficiency claims, which required direct measurement precisely because they did *not* generalize from architectural reasoning. The generalization claim is scoped deliberately: both backbones are pure multi-head self-attention `VisionTransformer` instances of different widths and depths, not different architecture families – whether the ranking survives a hybrid conv+attention backbone remains untested, since relevancy propagation is not defined for EfficientViT-M0's architecture (Section 6, External validity).

4.9 System-integration: does combining two uncertainty signals help?

Table 9. Selective-prediction risk-coverage AUC (lower is better): confidence alone vs. conformal set size alone vs. combined.

Condition	Confidence only	Set size only	Combined
In-distribution	**0.0050**	0.0834	0.0173
Herlev (moderate shift)	**0.1190**	0.3281	0.1793
Mendeley LBC (severe shift)	0.3244	0.4395	**0.2855**

Composition is not a uniform win. In distribution and under moderate shift, confidence alone is already the best individual signal, and combining it with the noisier discrete conformal-set-size signal dilutes it. Under severe shift, however, the combined signal is the best of the three, beating confidence alone. This mirrors the shape of the uncertainty-method finding in Section 4.5–4.6 (no single choice wins everywhere), now extended to combinations of signals.

4.10 Resource efficiency and deployment robustness

Across the 12-model registry, EfficientNet-Lite0 (3.38M params, 0.77G FLOPs) has more than 4× ResNet-50's CPU latency (137.7ms vs. 32.0ms) despite ResNet-50 having roughly 10× the FLOPs. Even EfficientViT-M0, the overall winner on params+FLOPs+accuracy (Section 4.1), is not the fastest model in wall-clock CPU terms: DeiT-Tiny (15.4ms) and LeViT-128 (17.5ms) are both faster despite 10–14× more FLOPs. EfficientViT-M0, evaluated under synthetic JPEG compression, Gaussian blur, and brightness/contrast shift, degrades gracefully under all three, but Gaussian blur is by far the most damaging – costing 32.8 points of balanced accuracy at its most severe tested level, roughly double JPEG's worst case (15.5 points) or brightness/contrast's (18.1 points). A synthetic OOD sanity check shows mean confidence dropping from 0.985 (real in-distribution images)

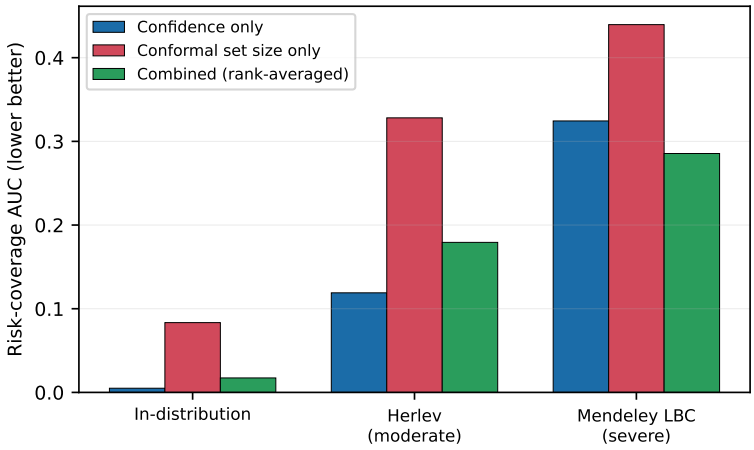


Fig. 6. Risk-coverage AUC for confidence alone, conformal set size alone, and the rank-averaged combination, at three conditions. Combining hurts in-distribution and under moderate shift but helps under severe shift.

to 0.743 (OOD) – real and meaningful, but far above the 0.20 chance level for this 5-class task. On CPU, a bare DeiT-Tiny forward pass has a median latency of 49.6ms; a prediction that also returns a relevancy-propagation explanation costs 164.1ms – about 3.3× the bare-prediction cost.

5 Discussion

Seven findings, read together, argue against a single verification strategy applied uniformly to a trustworthy medical AI pipeline. On **efficiency and shift robustness** (Sections 4.1–4.4): "smaller is worse" is directly falsified in distribution; "the in-distribution winner is the safe choice" is directly falsified under shift, and replicates this reversal on a third, independently sourced dataset; "architecture family/lineage predicts robustness" is falsified by MobileViT-S and MobileViTv2 sharing a name and lineage while producing a catastrophic failure and a top performance on the same evaluation; and, visible only once a third shift dataset and a seed-variance check are both added, even a robustness pattern that survives two independent shift datasets, and even the specific model used to illustrate the paper's own central claim, are not guaranteed to survive a third shift dataset or a different random seed respectively. A reader who stopped at two shift datasets and one seed would have concluded "CNNs are the robust family" and "EfficientViT-M0 is reliably the in-distribution-winner-becomes- shift-loser example"; both conclusions are one seed or one dataset away from being overturned.

On **uncertainty quantification** (Sections 4.5–4.6): a reader who evaluated the four methods only in distribution, or only under the milder of the two shifts

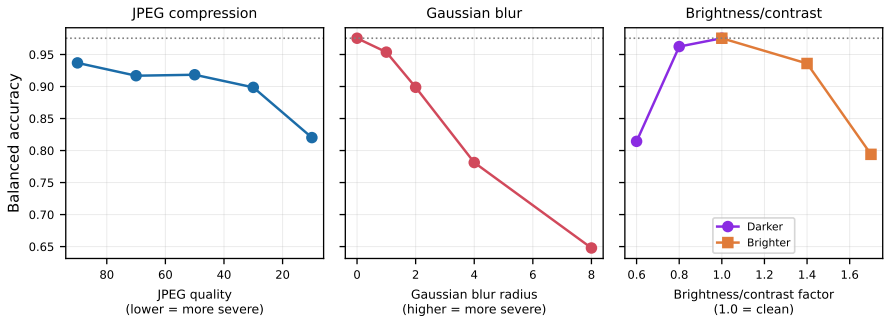


Fig. 7. Balanced accuracy vs. corruption severity for three synthetic degradations (EfficientViT-M0, SIPaKMeD test set). The dotted line marks the clean baseline (0.975). Gaussian blur causes the steepest decline.

tested, would conclude the deep ensemble is the best choice and MC Dropout is mediocre – the opposite of what the severe-shift evaluation shows in our original run. The significance test and repeated retraining add a second, sharper layer to this same warning: it was not enough for MC Dropout's severe-shift advantage to fail a bootstrap significance test against the ensemble – retraining the exact same method with the exact same seed produced a qualitatively different result (0.285 becoming 0.503), large enough to flip which method looks better at all. If a single fixed backbone choice cannot be assumed safe across shift conditions (Sections 4.1–4.4), and a single fixed uncertainty-method choice cannot be assumed safe across shift conditions either, this result adds a third, more uncomfortable layer: a single uncertainty-method *evaluation run* cannot be assumed to represent that method's behavior on a severe shift, even holding the seed fixed.

On **explainability, composition, and measured cost** (Sections 4.7–4.10): explainability method choice produced the cleanest, most unambiguous result in the entire paper – relevancy propagation wins every metric, with no condition-dependent flip, and the ranking replicates exactly on a second backbone – while uncertainty method choice, uncertainty-signal composition, and architecture-family robustness patterns all produced conditional, no-single-winner, or seed-dependent results. This asymmetry suggests that for this task and backbone family, picking the right explainability method is a more reliably beneficial, one-time engineering decision than picking (or combining) the right uncertainty-quantification method or trusting a single seed's robustness ranking, where the honest answer is "it depends, and needs re-testing." Every efficiency and robustness claim, meanwhile, required direct measurement to be trusted: FLOP count did not predict CPU latency, and OOD confidence dropped only partially, not to the clean near-chance level a reader might assume from an "OOD detection" framing.

Together, these seven findings do not all fail to compose the same way – that is the point. Some properties (explainability method choice) are robust once tested with reasonable care. Some (uncertainty method choice, signal composition) are conditional on an operating regime that must be identified before trusting any single-condition evaluation. Some (architecture-family robustness patterns, and even which specific model best illustrates a general claim) are not safely generalized beyond the exact seed and dataset used to establish them, and require deliberate, repeated re-testing to distinguish a real pattern from a one-run artifact. A system designer who validates one of these properties once and assumes the others behave the same way is making an assumption this data does not support, at three different levels: across models, across shift datasets, and across random seeds of the identical model and recipe.

6 Threats to Validity

Internal validity. Each headline model in Sections 4.1–4.3 was trained once (seed 13); Section 4.4 measured seed-to-seed variance directly across all 12 models at 2 further seeds (17, 29) rather than leaving it as a caveat, and found some specific single-model, single-dataset claims (the Mendeley LBC CNN-vs-transformer split, EfficientViT-M0 as the in-distribution-winner-becomes-shift-loser example) are seed-fragile, while others (Herlev's inverse pattern, MobileNetV2's APCEM win) are seed-stable – Section 4.4 gives the full breakdown. No trained-weight checkpoints persist across sessions, so results could not be re-verified from the original weights without full retraining; the reported numbers are the direct, unedited output of the released training and evaluation code, which mitigates but does not eliminate this threat. For the uncertainty-quantification comparison (Sections 4.5–4.6), this threat is no longer hypothetical: we tested it directly by retraining MC Dropout three more times (seed 13 again, then genuinely different seeds 17 and 29), all under the identical dropout rate, epoch budget, and manifest used for Table 5. The four Mendeley LBC risk-coverage AUC values – 0.285, 0.503, 0.358, 0.359 – span 0.218, with the two seed-17/29 runs landing in between the two nominally-seed-13 runs rather than clustering near either. This swing erases the reported severe-shift advantage entirely and makes at least one retrained run statistically indistinguishable from the deep ensemble (paired bootstrap 95% CI for the difference $[-0.066, 0.010]$, one-sided $p = 0.929$; Section 4.6). The deep ensemble's 5 members use one fixed set of seeds, reused unchanged for this significance test; seed-to-seed ensemble variance was not separately measured. For the explainability comparison (Sections 4.7–4.8), the 50-image evaluation sample is one fixed stratified sample (10 per class); a different sample could shift the exact deletion/insertion/nucleus-concentration numbers, though the margin by which relevancy propagation wins on all three metrics, on both backbones, is large enough that a resampling-driven reversal of the ranking is unlikely. The OOD sanity check's 25-image sample (Section 4.10) is smaller still, and its qualitative conclusion (confidence drops but not to chance) should be treated as suggestive rather than a precise estimate for this reason. CPU la-

tency figures (Section 4.10) were measured on one shared consumer laptop rather than isolated benchmark hardware; a ∼30% run-to-run swing was observed for the same model, so the exact 3.3× explanation-cost multiplier should be read as approximate, not a precise constant, though relative ordering across models was stable.

External validity. All four datasets are drawn from a small number of source institutions (SIPaKMeD, Herlev, Mendeley LBC, and APCEM together represent four geographic/clinical sources), and generalization to other scanners, staining protocols, or populations beyond these four is not established. The three shift conditions tested (Herlev, Mendeley LBC, APCEM) are real cross-institution shifts, not a systematic sweep across shift types (e.g., scanner resolution, stain intensity, patient demographics varied independently), so the finding that no static model property predicts shift robustness is demonstrated for these three shifts specifically, not proven for shift in general. The APCEM result itself is direct evidence for why this caveat matters in practice, not just in principle: the CNN-family robustness pattern that held across the first two shift datasets did not survive the third, so a reader should not assume the specific findings reported here (e.g., which individual model is most robust) transfer to a fourth, untested shift condition either – only the higher-level conclusion, that no static property reliably predicts shift robustness, is expected to generalize. The uncertainty-quantification comparison (Sections 4.5–4.6) uses a single backbone (EfficientViT-M0) and two shift conditions; whether the same rank inversion (ensemble best under moderate shift, MC Dropout best under severe shift) would recur with a different backbone, a third shift condition, or a continuous rather than two-point severity scale is untested and is the most direct follow-up this result motivates. The explainability ranking (Sections 4.7–4.8) was re-tested on a second backbone, ViT-Small, and replicated exactly, but both DeiT-Tiny and ViT-Small are standard multi-head self-attention transformers of the same architectural family, chosen specifically because relevancy propagation is defined for them and not for EfficientViT-M0's hybrid design; this closes the single-backbone gap only within that family. The selective-prediction composition result and the degradation/OOD findings (Sections 4.9–4.10) were not re-tested on ViT-Small and still use DeiT-Tiny alone. Whether any of the explainability, composition, or degradation/OOD results transfer to a hybrid conv+attention backbone, or to a different medical imaging task entirely, remains untested.

Construct validity. Balanced accuracy, macro F1, and AUROC (Sections 4.1–4.4) are standard, well-understood metrics for imbalanced multi-class classification, reducing the risk that the efficiency or robustness claims are artifacts of an unusual metric choice. Measured CPU/GPU latency is a single-run, single-batch-size figure per model and is not the focus of the paper's central efficiency/robustness claims, so its precision is secondary to those findings. Risk-coverage AUC and selective accuracy at fixed coverage (Sections 4.5–4.6) are standard selective-prediction metrics with clear operational meaning, reducing the risk that the rank inversion is an artifact of an unusual metric choice. Evidential deep learning's calibration numbers are reported without the

post-hoc correction its raw outputs would need for a fully fair comparison to the other three methods' already-calibrated numbers – a known asymmetry in how the four methods are scored, stated explicitly rather than smoothed over. Deletion/insertion AUC and nucleus concentration ratio (Sections 4.7–4.8) are both well-established or directly interpretable metrics, reducing the risk that the explainability ranking is an artifact of an unusual scoring choice; using SIPaKMeD's own expert-marked nucleus polygons as ground truth, rather than a segmentation model's derived output, specifically avoids compounding this evaluation with a second model's own errors. No clinical or expert-review validation of explanation localization fidelity was performed anywhere in this paper – agreement with a nucleus boundary is not equivalent to a cytotechnologist judging an explanation clinically meaningful, a distinction stated explicitly rather than implied away.

7 Limitations

Three external shift datasets, all drawn from a small number of source institutions; the shift conditions tested here are real but not exhaustive of the shift a deployed system might encounter, and the architecture-family pattern found stable across the first two shift datasets did not survive the third, so any single-dataset robustness ranking reported here should be read as specific to that dataset rather than as a general robustness ordering. Each Section 4.1–4.3 headline result is a single train/val/calibration/test split (seed 13); results are not averaged over multiple seeds by default, though seed-to-seed variance was measured directly (Section 4.4, Section 6 Internal validity) across all 12 models at 2 further seeds rather than left unmeasured, and that check found some single-model and single-dataset claims are seed-specific rather than general, so individual rows and close-ranked comparisons in these tables should be read with that caution; the paper's higher-level conclusions do not rest on any single seed-fragile comparison. No trained-weight checkpoints persist; reproducing any single row of these tables means retraining from scratch. The uncertainty-quantification comparison (Sections 4.5–4.6) uses a single backbone (EfficientViT-M0); whether the reported inversion (ensemble best under moderate shift, MC Dropout best under severe shift) holds for other architectures is untested, and its own severe-shift ranking was directly shown not to reproduce under retraining (the paper's own central caution here). "Moderate" and "severe" shift are descriptive labels based on the accuracy drop observed, not independently validated shift-magnitude measurements. Evidential deep learning's calibration numbers are reported without the post-hoc correction its raw outputs would need for a fully fair comparison. Deep ensemble results depend on the specific 5 seeds used; seed-to-seed ensemble variance was not separately measured. The explainability and system-integration results (Sections 4.7–4.10) use DeiT-Tiny, not the efficiency benchmark's overall winner, EfficientViT-M0, since relevancy propagation is not defined for the latter; the explainability ranking specifically was confirmed on one additional same-family backbone (ViT-Small), but whether any of these results transfer to

a hybrid conv+attention backbone remains untested. CPU latency measured on one shared consumer laptop, not dedicated hardware; a ∼30% run-to-run swing was observed for the same model, though relative ordering across models was stable. The OOD sanity check uses a small, synthetic sample (25 images) rather than a natural out-of-domain dataset. No clinical or expert-review validation of explanation localization fidelity was performed anywhere in this paper.

8 Conclusion

Testing five independently-varying properties of one trustworthy cervical cytology screening pipeline, side by side rather than one at a time, does not produce one story – it produces three. Some properties resolve cleanly and hold up under scrutiny: relevancy propagation is unambiguously the most faithful of three explainability methods tested, on both behavioral faithfulness and real expert-annotated localization ground truth, and this ranking replicates exactly on a second, independently- trained backbone. Some properties resolve conditionally, flipping direction with the operating regime: composing two uncertainty signals for selective prediction helps only where the better individual signal was already failing, and the uncertainty-quantification method that wins under moderate distribution shift is not the method that wins under severe shift. And some properties that look stable turn out to be artifacts of a single seed, dataset, or trained instance once tested again under identical conditions: a CNN-family robustness pattern that survived two external shift datasets collapsed on a third; this paper's own headline illustration of an in-distribution winner becoming a shift-robustness loser does not reliably reproduce across random seeds of the identical model; and an uncertainty method's reported advantage under severe shift vanished on retraining with the identical seed and recipe, with four independent runs spanning a wider gap than separates any two different methods. Twelve models across three architecture families also show, contrary to the usual efficiency-accuracy trade-off framing, that parameter-efficient vision transformers cost nothing here – the smallest models in the registry are also the most accurate in distribution – and resource-efficiency claims for lightweight architectures resolve only under direct measurement, in both directions: FLOP-efficient does not mean latency-efficient, and "confidence drops on OOD input" does not mean "confidence reaches a safe abstention level." No single verification strategy – pick the best method once, assume a pattern generalizes across datasets, trust that a fixed seed represents a method's behavior, or trust an architectural efficiency claim without measuring it – would have correctly characterized all five properties of this one pipeline tested here; each required its own test, repeated where a single run could not answer the question, on its own terms.

References

1. Plissiti, M.E., Dimitrakopoulos, P., Sfikas, G., Nikou, C., Krikoni, O., Charchanti, A.: SIPAKMED: A new dataset for feature and image based classification of normal

and pathological cervical cells in Pap smear images. In: 2018 IEEE International Conference on Image Processing (ICIP), pp. 3144–3148 (2018)
2. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: CVPR (2016)
3. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.C.: MobileNetV2: Inverted residuals and linear bottlenecks. In: CVPR (2018)
4. Tan, M., Le, Q.: EfficientNet: Rethinking model scaling for convolutional neural networks. In: ICML (2019)
5. Dosovitskiy, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: ICLR (2021)
6. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin Transformer: Hierarchical vision transformer using shifted windows. In: ICCV (2021)
7. Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., Jégou, H.: Training data-efficient image transformers & distillation through attention. In: ICML (2021)
8. Graham, B., El-Nouby, A., Touvron, H., Stock, P., Joulin, A., Jégou, H., Douze, M.: LeViT: a vision transformer in ConvNet's clothing for faster inference. In: ICCV (2021)
9. Wu, K., et al.: TinyViT: Fast pretraining distillation for small vision transformers. In: ECCV (2022)
10. Mehta, S., Rastegari, M.: MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer. In: ICLR (2022)
11. Mehta, S., Rastegari, M.: Separable self-attention for mobile vision transformers. arXiv preprint (2022)
12. Cai, H., Li, J., Hu, M., Gan, C., Han, S.: EfficientViT: Multi-scale linear attention for high-resolution dense prediction. In: ICCV (2023)
13. Liu, X., Peng, H., Zheng, N., Yang, Y., Hu, H., Yuan, Y.: EfficientViT: Memory efficient vision transformer with cascaded group attention. In: CVPR (2023)
14. Jantzen, J., Norup, J., Dounias, G., Bjerregaard, B.: Pap-smear benchmark data for pattern classification. In: Proceedings of Nature Inspired Smart Information Systems (NiSIS), pp. 1–9 (2005)
15. Hussain, E., Mahanta, L.B., Borah, H., Das, C.R.: Liquid based-cytology Pap smear dataset for automated multi-class diagnosis of pre-cancerous and cervical cancer lesions. Data in Brief 30, 105589 (2020)
16. Aswini, H.: APCEM-Enhanced Cervical Cytology Image Dataset (Version 2) [Data set]. Mendeley Data (2026). https://doi.org/10.17632/snzb5g6ddy.2
17. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: ICML (2017)
18. Gal, Y., Ghahramani, Z.: Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In: ICML (2016)
19. Lakshminarayanan, B., Pritzel, A., Blundell, C.: Simple and scalable predictive uncertainty estimation using deep ensembles. In: NeurIPS (2017)
20. Sensoy, M., Kaplan, L., Kandemir, M.: Evidential deep learning to quantify classification uncertainty. In: NeurIPS (2018)
21. Chefer, H., Gur, S., Wolf, L.: Transformer interpretability beyond attention visualization. In: CVPR (2021)
22. Abnar, S., Zuidema, W.: Quantifying attention flow in transformers. In: ACL (2020)
23. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: ICCV (2017)

24. Petsiuk, V., Das, A., Saenko, K.: RISE: Randomized input sampling for explanation of black-box models. In: BMVC (2018)
25. Mohiuddin, A.B., Rahman, M.M., et al.: Accelerated and accurate cervical cancer diagnosis using a novel stacking ensemble method with explainable AI. In: International Conference on Quantum Photonics, Artificial Intelligence (2025)
26. Rahman, M.M., et al.: MaizeFormerX: a lightweight vision transformer with cross-scale attention for explainable maize leaf disease diagnosis. Scientific Reports (2026)
27. Gony, M.N., Shuvra, S.M.K., Poudel, R., Dhakal, K., Rahman, M.M., et al.: Dual-view semi-supervised retinal disease recognition via chromatic structural decomposition and cross-view contrastive learning. In: International Conference on Power, Electronics, Communications (2026)
28. Ahmed, M., Ahmed, F., Mow, S.M., Taha, P.A., Barua, S., Rahman, M.M., et al.: An explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support. medRxiv (2026)